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Palaeomagnetic results from the ca. 1130 Ma Borgmassivet intrusions in the Ahlmannryggen region of Dronning Maud Land, Antarctica, and tectonic implications

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Abstract

Detailed palaeomagnetic and rock magnetic analyses provide improved palaeomagnetic results from 23 sites in the Borgmassivet intrusions in the Ahlmannryggen region of Dronning Maud Land, East Antarctica. These intrusions are of similar age to their host, the ca. 1130 Ma Ritscherflya Supergroup (RSG). A mean direction of $D=235.4°$, $I=-7.6°$ with $k=45.9$ and $\alpha_{95}=4.5°$ was obtained from this study. When combined with previously reported results from 11 sites in the same region, including sites from the Ritscherflya Supergroup, it gives an overall mean direction for 34 sites from the igneous suite with $D=236.5°$, $I=-3.6°$, $k=27.9$ and $\alpha_{95}=4.8°$. Isothermal remanent magnetization (IRM) experiments on several specimens suggest magnetite or titanomagnetite as the primary remanence carrier, while high temperature magnetic susceptibility experiments indicate the presence of single domain particles. These observations, together with field evidence and the high coercivities and unblocking temperatures, support a primary origin for the observed characteristic remanence. The Borgmassivet palaeomagnetic pole lies at 54.5°E, 8.3°N with $A_{95}=3.3°$. If Antarctica is moved to its Gondwanan position adjacent to southeast Africa, the Borgmassivet pole (BM) coincides with that of the African well-established, well-dated (1100 Ma) Umkondo Large Igneous Province pole, supporting the hypothesis that the Grunehogna craton of Dronning Maud Land was part of the Kalahari craton of southern Africa at ca. 1100 Ma.
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1. Introduction

Geological similarities between rocks in the Grunehogna craton of Dronning Maud Land, East Antarctica (Fig. 1a), and those in the Kalahari craton of southern Africa have led to the suggestion that the two regions were in conjunction in the Mesoproterozoic (e.g., Groenewald et al., 1995). The Ritscherflya

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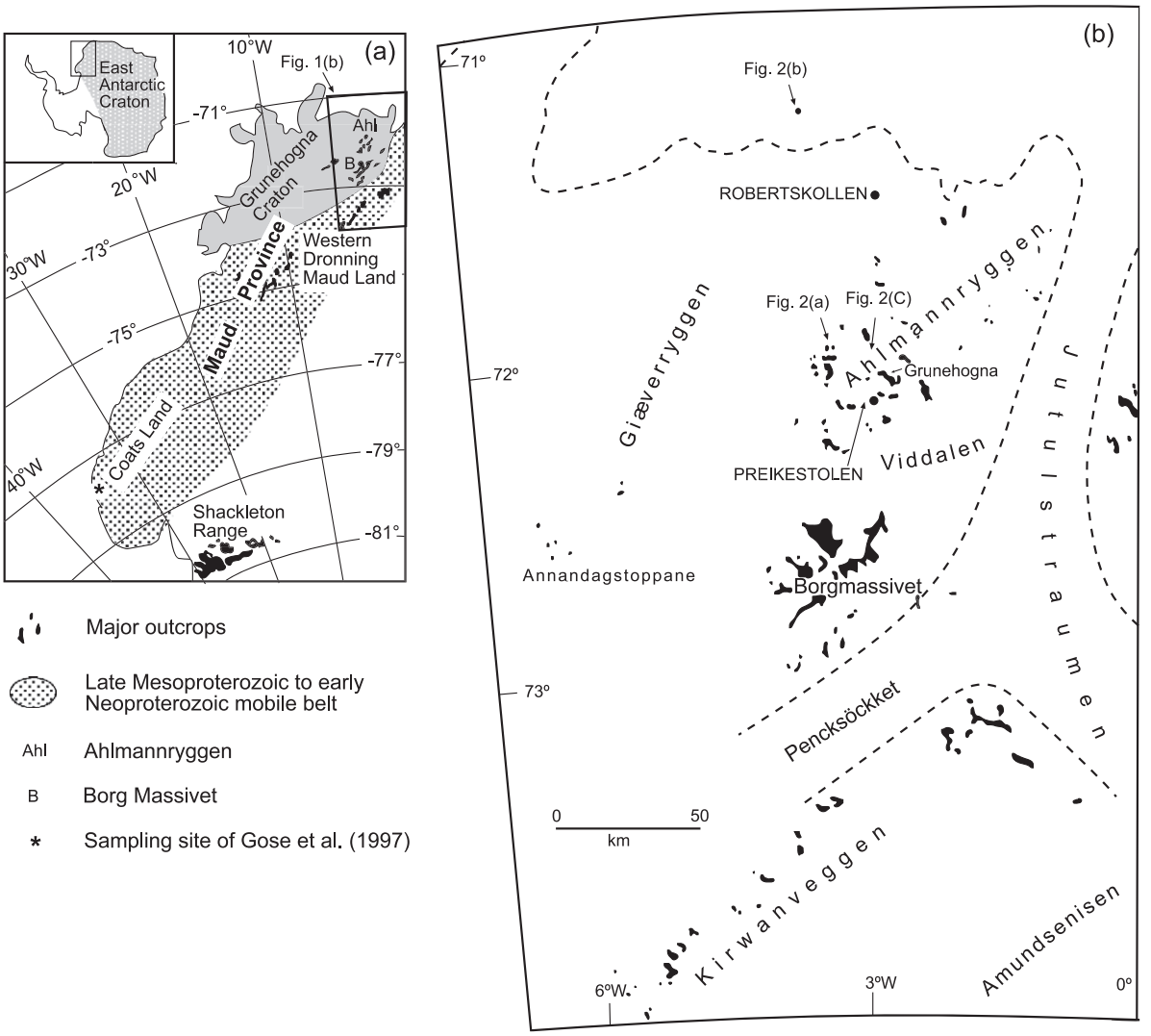


Fig. 1. Geological framework in western Dronning Maud Land (a) and locations of the study region (b).

Supergroup (RSG) of Dronning Maud Land is taken as a possible correlative of the Umkondo Group of eastern Zimbabwe with both being intruded extensively by mafic rocks before consolidation was complete. The Umkondo dolerites have recently been dated using both conventional and SHRIMP U–Pb zircon methods as close to 1100 Ma (Hanson et al., 1998; Wingate, 2001). The age of the Borgmassivet intrusions into the RSG, although less well established, is indicated by various studies also to be close to 1100 Ma (see discussion below).

The possible connection between the Coats Land and western Dronning Maud Land of East Antarctica (the Maud Province, e.g., Fitzsimons, 2000) and Kalahari is important because the Grenvillian orogenic belt in the Maud Province was once compared with the classic Grenville Belt in Laurentia and used as a key evidence for the SWEAT hypothesis (e.g., Moores, 1991; Hoffman, 1991; Dalziel, 1991). However, subsequent palaeomagnetic analysis of ca. 1110 Ma igneous rocks from Coats Land (Fig. 1a) by Gose et al. (1997) revealed that either East Antarctica was

not where it was suggested to be in the SWEAT connection, or that the Maud Province was not part of the East Antarctica craton during late Mesoproterozoic. Gose et al. (1997) preferred the latter interpretation because of geological similarities between the Grunehogna craton in the Maud province and the Kalahari craton, and the apparent agreement between their ca. 1100 Ma palaeomagnetic poles. This interpretation has been widely accepted by later workers (e.g., Dalziel, 1997; Jacobs et al., 1998; Fitzsimons, 2000; Powell et al., 2001).

In order to further verify this interpretation, high-quality palaeomagnetic results from both the Maud Province and Kalahari are required. Reliable palaeomagnetic poles are available from rocks of the Umkondo Large Igneous Province in eastern Zimbabwe of the Kalahari craton (Briden et al., 1979; Onstott et al., 1986; Hargraves et al., 1994). However, published results for the Maud Province are only available from Coats Land (Gose et al., 1997), which is within the late Mesoproterozoic–early Neoproterozoic mobile belt (e.g., Fitzsimons, 2000; Golynsky and Jacobs, 2001) and is over 600 km away from the Grunehogna craton (Fig. 1a). The purpose of this paper is to present detailed palaeomagnetic results from part of a larger collection of Hodgkinson (1989), which, together with results from the Umkondo Large Igneous Province in eastern Zimbabwe, provides further evidence for the conjunction of the Grunehogna and Kalahari cratons some 1100 Ma ago.

2. Geology and geochronology

The Grunehogna craton of Dronning Maud Land lies to the west of the Jutulstraumen glacier and the Pencksöckket depression (Fig. 1b). Within the craton there are Archaean granites at Annandagstoppane while a Mesoproterozoic volcanisedimentary sequence, the RSG, occurs throughout the Borgmassivet and Ahlmannryggen regions. The sediments have been described by Ferreira (1986) as ‘‘shallow marine, tidal flat, braided stream and alluvial fan deposits’’. Within the Hogfonna and Raudberget Formations of the RSG, there are reddish-brown sandstones and quartzites. The RSG is capped by volcanogenic formations, which include the basalts of the Fassetfjellet Formation. The RSG is almost flat-lying (except on the edge of the Jutulstraumen) and has undergone only lower greenschist facies regional metamorphism (Wolmarans and Kent, 1982).

The RSG has been intruded extensively by the Borgmassivet intrusions, which are mafic to ultramafic in composition. The intrusions occur as sills up to 400 m thick (e.g., the Grunehogna and Kullen sills at Grunehogna, which intrude the Hogfonna and Schumacherfjellet Formations) or as layered intrusions (e.g., the Robertskollen Suite). There is considerable evidence that the Grunehogna sill intruded both the Hogfonna and Schumacherfjellet Formations while they were still wet, whereas the slightly younger Kullen sill was emplaced in the still unconsolidated Hogfonna Formation (Krynauw et al., 1988).

A U–Pb date on zircons from a tuff within the RSG sediments gives an age of ca. 1135 Ma (unpublished data of M. Knoper and R. Tucker, reported in Powell et al., 2001). This is consistent with the U–Pb crystallisation age of 1130 ± 12 Ma from metavolcanic rocks in the banded gneisses of the Petermannketten area, east of the RSG (Jacobs et al., 1998), which have been interpreted as metamorphosed equivalents of the RSG (Powell et al., 2001). The Borgmassivet intrusions and the RSG are thus regarded tentatively to be ca. 1130 Ma.

3. Previous palaeomagnetic work

Hodgkinson (1989) collected samples from 54 sites, principally in the Borgmassivet intrusions with various rock types represented, throughout the Ahlmannryggen and Borgmassivet areas. Pilot specimens were treated progressively with alternating field (AF) demagnetization, but blanket AF demagnetization was applied to the bulk of the samples. Peters (1989) reported results from 11 sites from both lavas and Borgmassivet intrusions in the Ahlmannryggen region (Fig. 1b). Samples were subjected to both stepwise AF and thermal demagnetization. Consistent results were obtained, with samples from the lavas and intrusives yielding essentially the same result. However, neither of the two authors have published their results.

In this study, we re-visited 23 sites of Hodgkinson’s large collection, all from the Ahlmannryggen area. We performed more detailed palaeomagnetic and

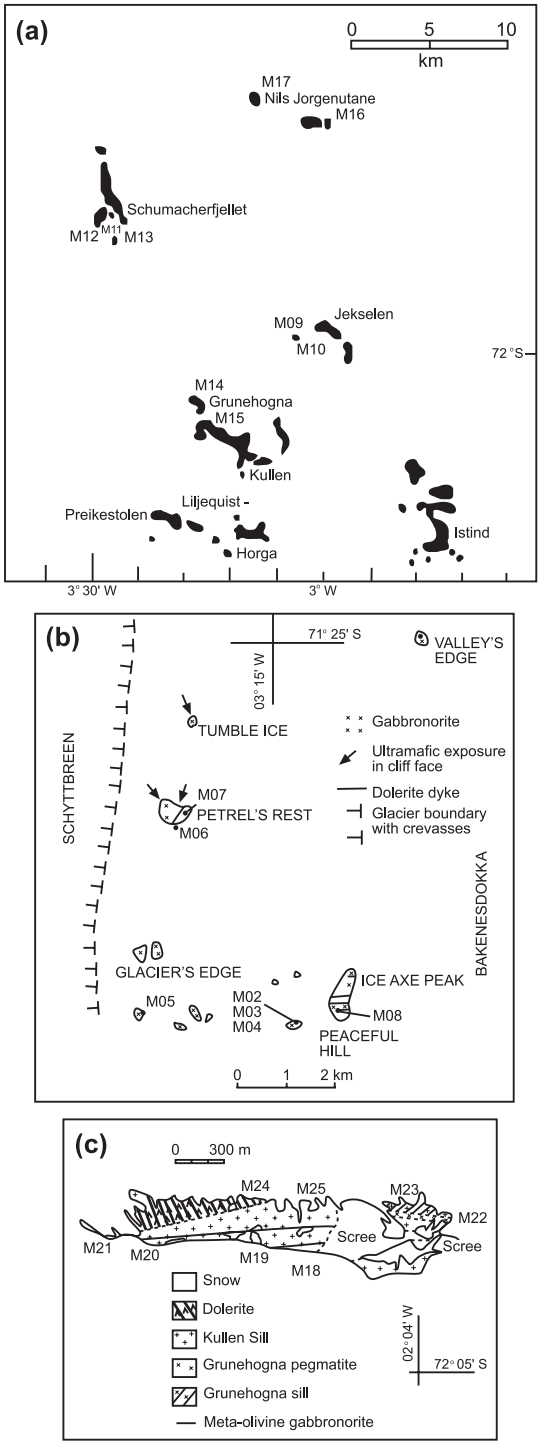


Fig. 2. Locations of sampling sites at (a) central Ahlmannryggen, (b) Robertskollen and (c) Preikestolen.

rock magnetic analyses on the samples, and report the results here.

4. Sampling

Core samples were collected from 25 sites in the Ahlmannryggen region during the 1987/1988 and 1988/1989 field seasons. While a variety of rock types are represented in the collection, the majority of the sites are in dolerites. Generally, four or five cores, each of 25 mm diameter, were obtained at each site using a hand-held petrol-driven drill lubricated with a water-soluble oil–ethanol mixture (to avoid freezing). Cores were oriented by means of a sun compass. Cores were cut into 25 or 22 mm long specimens depending on the magnetometer used for remanence

Table 1
Details of palaeomagnetic sites in Borgmassivet intrusions in the Ahlmannryggen region of Dronning Maud Land, Antarctica

Site	Site coordinates		Locations/ formation	Rock type	*N*
	Lat. (S)	Long. (W)			
M02	71.40	3.23	Robertskollen	cgr felsic zone, in gabbronorite	5
M03	71.40	3.23	Robertskollen	gabbronorite	5
M04	71.40	3.23	Robertskollen	cgr felsic zone, in gabbronorite	5
M05	71.40	3.23	Robertskollen	cgr felsic zone, in gabbronorite	5
M06	71.45	3.20	Robertskollen	gabbronorite	5
M07	71.45	3.30	Robertskollen	felsic zone, in gabbronorite	5
M09	71.99	2.61	Jekselen	dolerite	4
M10	72.01	2.54	Jekselen	dolerite	4
M11	71.93	2.96	Schumacherfjellet	dolerite	4
M12	71.93	2.96	Schumacherfjellet	dolerite	4
M13	71.93	2.94	Schumacherfjellet	dolerite	4
M14	72.03	2.80	Grunehogna	dolerite/syenite	4
M15	72.03	2.80	Grunehogna	cgr syenite	4
M16	71.87	2.61	Nils Jorgennutane	syenodiorite	4
M17	71.87	2.61	Nils Jorgennutane	diorite	4
M18	72.09	2.08	Preikestolen (KS)	dolerite	4
M19	72.09	2.08	Preikestolen (KS)	dolerite	4
M20	72.09	2.08	Preikestolen (KS)	dolerite	4
M21	72.09	2.08	Preikestolen (KS)	dolerite	4
M22	72.09	2.08	Preikestolen	GH pegmatite	5
M23	72.09	2.08	Preikestolen (GHS)	dolerite	4
M24	72.09	2.08	Preikestolen	dolerite	4
M25	72.09	2.08	Preikestolen	GH pegmatite	6

N—number of samples drilled, cgr—coarsegrained, KS—Kullen sill, GH(S)—Grunehogna (sill).

measurements. Site localities are shown in Fig. 2. It was not possible to cut specimens from the friable pieces of core from site M01, while results from M08, in an ultramafic dyke intruding into the gabbronorites of the Robertskollen Suite, suggest a much younger, probably Phanerozoic, age for the dyke. Table 1 gives details of the remaining 23 sites, including rock types.

5. Measurements

Hodgkinson (1989) measured the remanent magnetization of specimens with fluxgate spinner magnetometers at the former Geological Survey of South Africa (now the Council for Geoscience), Pretoria. AF demagnetisation was done there using apparatus similar to that described by McElhinny (1966b). The detailed work described herein was performed at the University of Zimbabwe in Harare, the Oxford University of the UK and the University of Western Australia in Perth. Measurements in Harare and Oxford were made using fluxgate spinner magnetometers, with AF demagnetisation done in Harare with apparatus as in McElhinny (1966b), while for thermal demagnetisation Magnetic Measurements MMTD instruments were used in all three laboratories. A cryogenic magnetometer was employed in Perth while isothermal remanent magnetization (IRM) and high temperature magnetic susceptibility ($K-T$) experiments were run in Oxford using a Molspin pulse magnetizer and a Kappa bridge, respectively. Generally, a pilot specimen from each site was subjected to progressive AF demagnetization in 5 or 10 mT steps. If specimens had already been blanket demagnetised, generally only at 10 or 20 mT, but occasionally at 35 or 40 mT (4 sites), they were demagnetised again at that field value before proceeding with progressive AF or thermal demagnetisation. At demagnetisation levels above 50 mT, there was often evidence of rotational remanent magnetisation (RRM) being introduced. This was minimized at each subsequent AF step by using the double demagnetisation technique of Hillhouse (1977). The maximum AF demagnetisation field depended on the extent of RRM introduced and was generally in the range of 60–100 mT. A second specimen was subjected to progressive thermal demagnetisation typically in 100 °C steps to 500 °C, and then much smaller steps depending on specimen response. The results from the pilots were used to determine which technique was most suited to that site. One specimen per sample was then demagnetised progressively using AF or thermal demagnetisation.

Results from each specimen were displayed on orthogonal vector plots and characteristics of linear sections determined by using LINEFIND (Kent et al., 1983) or, in Perth, by principal component analysis (Kirschvink, 1980). Site mean directions were calculated using Fisher (1953) statistics.

6. Rock magnetic results

IRM acquisition experiments were run on six specimens using magnetic fields up to 800 mT. Three

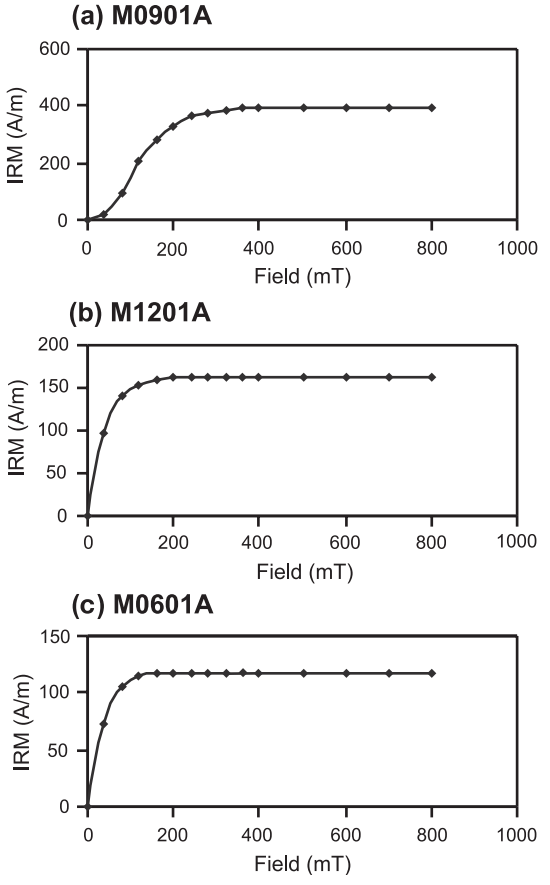


Fig. 3. Results of IRM experiments on specimens showing saturation in fields less than 300 mT.

of these (M0901A, M1201A and M2102B) were from sites in different areas in dolerite, while the rest (M0204A, M0601A and M0701B) were from sites in the gabbronorite at Robertskollen (Table 1). In all cases, a rapid rise in magnetization with increased applied field was observed (Fig. 3) with saturation achieved in fields generally by 200 mT (Fig. 3b and c) but with two saturating after 300 mT (Fig. 3a). Such behaviour is suggestive of magnetite or titanomagnetite being the dominant carriers of remanence, although a trace amount of pyrrhotite may exist in samples like M0901a (Fig. 3a; also see $K-T$ results below).

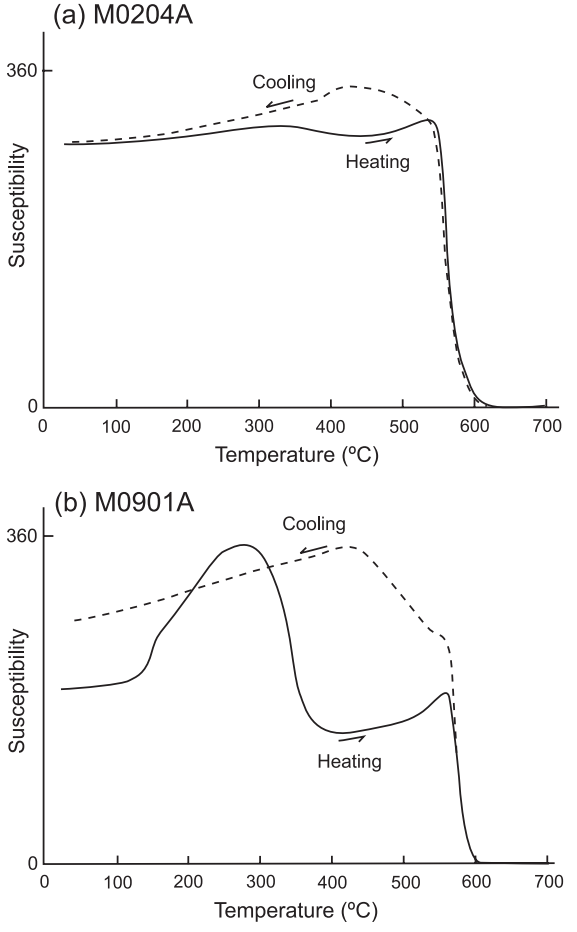


Fig. 4. Results of magnetic susceptibility-temperature experiments showing small Hopkinson peaks prior to reaching the Curie temperature of magnetite, and the possible presence of pyrrhotite.

$K-T$ experiments were run over the range of 25–700 °C on four specimens. Two of the specimens (M0204A and M1201A) show a gentle increase in susceptibility on heating to about 450 °C followed by a rapid decrease in susceptibility around 580 °C, with small but definite Hopkinson peaks (Fig. 4a). On cooling, the profile shows a broad maximum above but fairly close to the heating curve. The heating curve for M0901A (Fig. 4b) shows a prominent peak centred close to 300 °C and a susceptibility lower than the initial value after the peak, indicating the likely existence of some pyrrhotite. A sharp Hopkinson peak and a rapid decrease at about 580 °C indicate the existence of single-domain magnetite. The cooling curve illustrates the formation of new magnetite likely from the original pyrrhotite.

7. Magnetic remanence results and interpretation

The AF and thermal demagnetisation responses of pilot specimens varied from site to site as would be expected from a variety of rock types. Hodgkinson (1989) records very noisy AF responses from sites M15 and M22 and was unable to obtain meaningful results for these sites. Thus, thermal demagnetisation only was done on pilots from these sites. Characteristic remanence directions from AF and thermal demagnetisation pilots of individual sites never differ more than 11°, and are generally less than 5°. As thermal demagnetisation commonly gave less noisy data than AF demagnetisation, specimens from remaining samples from 20 sites were subjected to detailed thermal demagnetization, with 3 sites only subjected to AF demagnetisation.

In total, 115 specimens were subjected to progressive AF or thermal demagnetisation and, of these, only 5 (2 from M22) failed to produce a characteristic remanence, while orientation errors are suspected for 3 others. For five specimens (three from M22—a pegmatite), the characteristic remanence was determined by estimating an endpoint using a stereonet. Examples of the response to thermal and AF demagnetisation are given in Figs. 5 and 6. They show that the characteristic remanence is carried partially by magnetic grains of high coercivity and blocking temperatures close to the Curie temperature of magnetite (580 °C). With many specimens, a steep up-

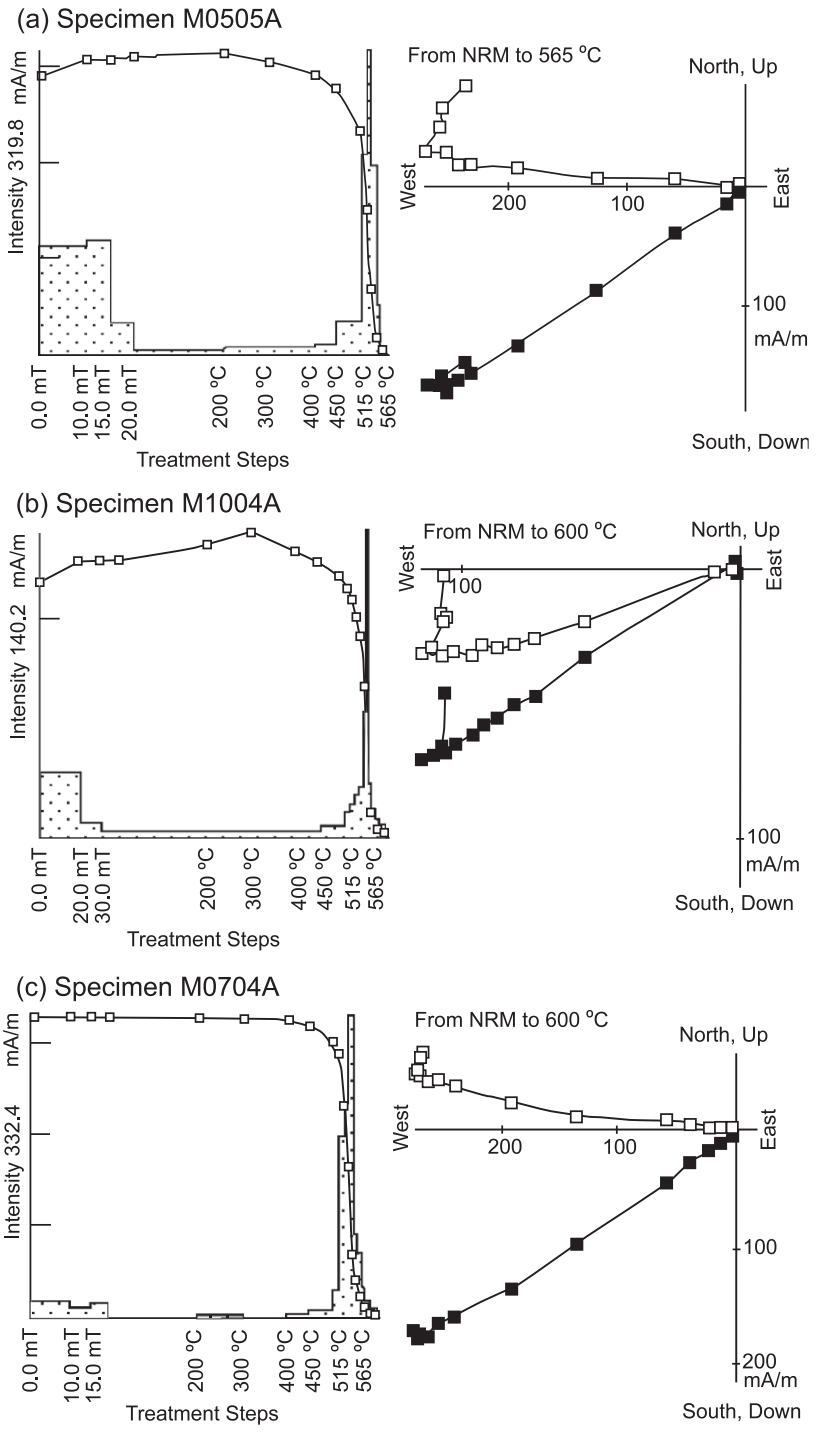


Fig. 5. Representative demagnetization curves where both AF and thermal methods have been used. In the orthogonal plots, filled squares are projections on horizontal planes and open squares are projections on vertical planes.

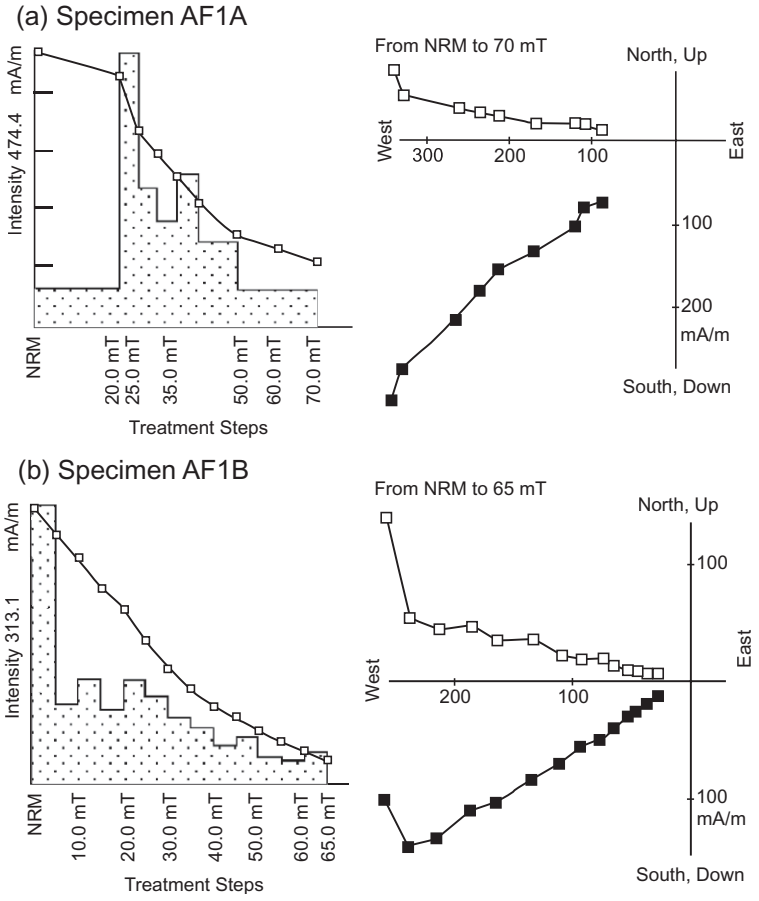


Fig. 6. Representative AF demagnetization curves. Projections are as in Fig. 5.

ward dipping component was removed in the early stages of AF or thermal demagnetization. For example, this component has a mean direction of $D=265°$, $I=-83°$ with $k=47$ for the four specimens in site M25. It is likely a viscous remanent magnetization was acquired in the Earth's present geomagnetic field.

Site-mean directions of characteristic remanence are given in Table 2.

Fig. 7 shows the site-mean directions of (a) natural remanent magnetization (NRM), (b) after blanket AF demagnetization only and (c) after stepwise AF or thermal demagnetization. Grouping of sample directions is not only improved for individual sites, but also in the overall distribution after stepwise demagnetization. Our new results (Fig. 7c) are also in good agreement with the data from 11 sites by Peters (1989) from the same region (Fig. 7d) (Table 3). Seven of Peters' (1989) 11 sites were from lava flows. Their agreement with data from the intrusive rocks further supports the interpretation of pene-contemporaneous nature of the lavas (and thus the RSG) and the intrusions. Table 3 gives an overall mean direction by combining our data with those of Peters' (1989), with $D=236.5°$, $I=-3.6°$, $k=27.9$ and $\alpha_{95}=4.8°$. The angular dispersion of directions is $15.3+3.2/-2.2°$, while the expected angular dispersion at the equator (Brock, 1971) is $18.9\pm4.0°$. Thus, there is no reason to believe that secular variation has not been averaged out.

Although no fold test or contact test was available for examining the timing of the remanent magnetization, we interpret the characteristic remanence as of primary origin based on (1) the dissimilarity of its direction from any expected Phanerozoic directions,

Table 2
Site mean directions of the primary remanence and associated VGPs for the Ahlmannryggen region of Dronning Maud Land, Antarctica

Site	GH					This study						VGP	
	n	blf (mT)	*D* (°)	*I* (°)	*k*	*N′*	*D* (°)	*I* (°)	*k*	α_{95} (°)	td or afd	Lat. (°S)	Long. (°E)
M02	10	20	236.1	−6.0	590	4	234.8	−7.2	1195	2.6	td	7.1	232.0
M03	11	20	239.9	−11.7	41	3	236.4	−7.0	736	4.5	td	6.8	233.6
M04	9	20	247.0	0.4	22	4	234.7	−7.5	413	4.5	td	7.0	231.9
M05	11	20	242.6	−5.2	2	3	236.3	−12.1	87	13.3	afd	4.3	232.8
M06	10	20	244.4	−9.5	14	3	232.1	−10.0	1299	3.4	td	6.4	229.1
M07	9	20	235.0	−10.7	938	5	236.6	−7.9	2868	1.4	td	6.3	233.6
M09	12	40	236.0	−11.8	45	4	225.2	8.0	120	8.4	td	16.5	225.0
M10	10	40	232.2	2.2	190	3	239.4	11.5	651	4.8	td	14.6	239.7
M11	12	20	245.8	−20.9	31	4	240.7	−19.8	23	19.4	td	−1.1	236.2
M12	12	20	233.3	−10.3	1192	4	231.8	−6.7	3000	1.7	afd	7.8	229.4
M13	10	35	231.6	16.8	314	4	236.6	15.9	130	8.1	td	17.6	237.2
M14	11	15	238.7	−9.2	595	4	240.1	−9.1	714	3.4	td	4.4	237.3
M15		0	59.1	−84.3		4	231.0	−31.8	13	26.6	td	−5.5	225.4
M16	12	30	226.7	−17.9	18	3	225.0	−24.9	19	28.9	td	0	220.9
M17	12	20	230.2	−17.7	635	4	232.8	−16.0	264	5.7	td	2.9	229.5
M18	12	35	236.5	8.9	109	4	241.8	−0.2	42	14.3	td	8.3	240.9
M19	16	15	257.8	17.8	50	3	235.3	−5.0	17	31.3	td	7.7	233.9
M20	14	20	238.0	3.6	307	4	232.5	0.0	364	4.8	td	10.8	231.8
M21	14	20	226.2	7.5	208	4	234.8	−2.9	723	3.4	td	8.8	233.7
M22		0	27.7	−83.1		3	251.9	0.1	135	10.6	td	5.5	250.7
M23	12	15	234.5	−7.8	500	4	235.6	−9.3	220	6.2	td	5.5	233.6
M24	10	0	240.2	−24.5	224	4	239.0	−11.0	137	7.9	afd	3.8	236.7
M25	4	10/20	169.1	−52.4	8	4	230.3	−9.4	165	13.6	td	6.7	228.5

Results from Hodgkinson (1989) are shown under the heading GH; *n* is the number of specimens blanket demagnetised (blf) in the given field; *D* and *I* are declination and inclination, respectively, of the site mean direction, while *k* is Fisher's (1953) precision parameter. td = thermal demagnetization and afd = alternating-field demagnetization. *N′* is the number of samples in the analysis; α_{95} is the semiangle of the cone of 95% confidence about the mean direction.

(2) the lack of any deformation in the intrusives and the RSG, (3) the rock magnetic results that suggest single-domain magnetite/titanomagnetite as the main magnetic carrier in many sites. Many specimens contain grains with coercivities greater than 70 mT (or even 100 mT) and blocking temperatures only just below the Curie temperature of magnetite (i.e., 580 °C). This would require a similar temperature to thermally remagnetise the mineral grains (e.g., Pullaiah et al., 1975). The virtually nonmetamorphosed rocks in the study region do not support the existence of such event. The lack of regional tilting, as shown by the flat-laying nature of the RSG, means that there is no palaeohorizontal ambiguity with the remanence direction.

The combined 34 sites listed in Table 3 give a (south) palaeomagnetic pole (BM) at 54.5°E, 8.3°N with a radius of 95% confidence (A_{95}) of 3.3°. This pole lies a few degrees east of the Horn of Africa. If Antarctica is rotated into the Gondwanan position using the rotation parameters given in Table 1 of Powell and Li (1994), the pole position becomes 33.3°E, 62.3°N.

8. Tectonic implications

Fig. 8 plots both the Borgmassivet pole (BM) pole from the Borgmassivet (representing the Grunehogna craton), and the CL pole from Coats Land (Gose et al., 1997) in present African coordinates, using the Powell and Li (1994) rotation parameters. It is apparent that the BM pole agrees well with the UMK pole from the Umkondo igneous suite in eastern Kalahari in this reconstruction. This appears to support geological correlation between the Gru-

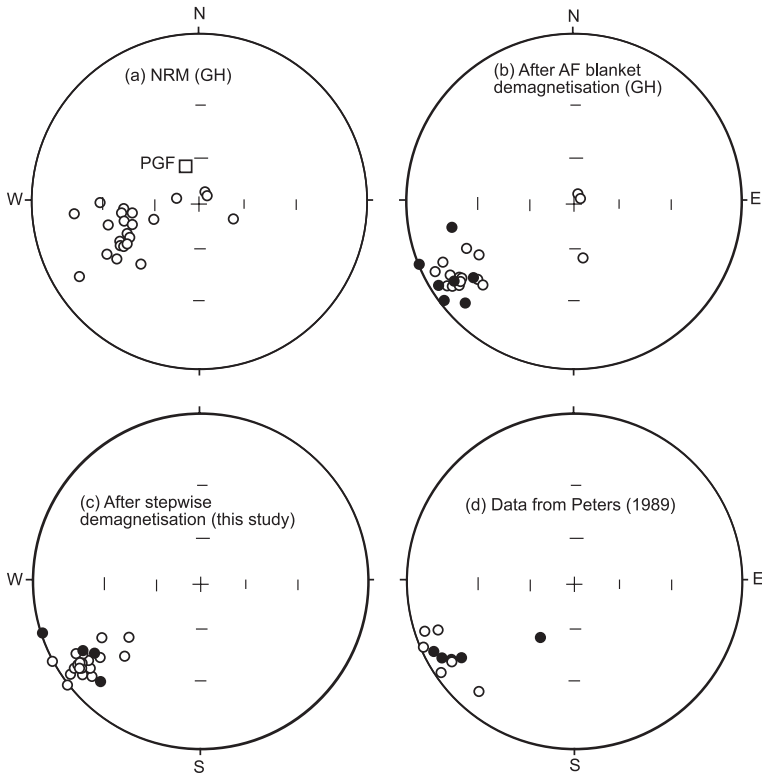


Fig. 7. Equal-angle stereographic projections showing site mean directions: (a) directions of site-mean NRM (Hodgkinson, 1989), (b) after blanket demagnetization (Hodgkinson, 1989), (c) results after stepwise demagnetization (this study) and (d) results of Peters (1989). Open (solid) circles denote negative (positive) inclinations. PGF = present geomagnetic field direction.

nehogna and Kalahari cratons, and between the RSG in the Maud Province and the Umkondo Group in eastern Kalahari (e.g., Groenewald et al., 1995).

The Umkondo Group of eastern Zimbabwe and the western parts of Mozambique is similar in several respects to the RSG in western Dronning Maud Land. There are arenaceous, argillaceous and calcareous formations resting unconformably on Archaean basement with the Umkondo lavas capping the Group. Dolerite sills up to 300 m thick intrude the Group and there is again some evidence of intrusion into only partially lithified sediments (Munyanyiwa, 1999). The Group is strongly deformed in the immediate border region, but, within a few kilometres, it is undeformed and dips only shallowly to the southeast. The dolerites are metamorphosed only to lowermost greenschist facies (Munyanyiwa, 1999).

Table 3
Mean directions of the characteristic remanence in circa 1100 Ma igneous rocks in the Ahlmannryggen region, Antarctica

Study	N	D (°)	I (°)	k	α_{95} (°)
A. Hodgkinson (1989) (blanket AF demagnetization only)	21	235.9	− 7.6	16.2	8.2
B. This study (stepwise demagnetization)	23	235.4	− 7.6	45.9	4.5
C. Peters (1989) (stepwise demagnetization)	11	238.7	4.0	17.8	11.1
B and C combined	34	236.5	− 3.6	27.9	4.8

N is the number of sites. Other symbols are defined in Table 2.

The Umkondo Group dolerites (UD) were first differentiated from the older Mashonaland dolerites (ca. 1830 Ma: Wingate and Jones, unpublished results) by McElhinny and Opdyke (1964); their palaeomagnetic remanence directions are approximately 90° apart from those of the UD. The Umkondo lavas were soon shown to be the extrusive equivalents of the UD

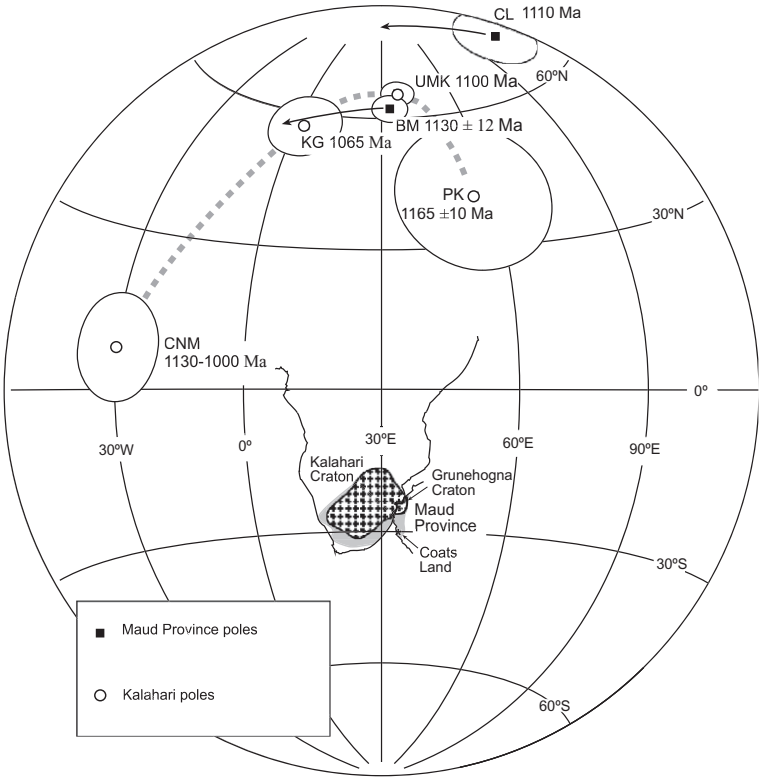


Fig. 8. Comparison of palaeomagnetic pole positions from the Maud Province with those from the Kalahari craton after the Maud Province (and its poles) is rotated to Africa according to the Gondwanan reconstruction parameters (7.09°N, 144.76°E, clockwise rotation by 56.34°) given by Powell and Li (1994). Arrows indicate shifting of pole positions if the Maud Province is rotated further by 20° counterclockwise around a local vertical axis. Maud Province poles: BM—Borgmassivet igneous suit (this study and Peters, 1989), CL—Coats Land (Gose et al., 1997). Kalahari craton poles: PK—Premier Kimberlite (Powell et al., 2001), UMK—Umkondo Igneous Province (Hargraves et al., 1994; Powell et al., 2001), KG—Kalkpunt Formation of the Koras Group (Briden et al., 1979), CNM—Central Namaqua metamorphic rocks (Onstott et al., 1986). Dotted line shows Kalahari APWP. More details of the Kalahari poles can be found in Table 1 of Powell et al. (2001). Map produced using GMAP (Torsvik and Smethurst, 1999) on Schmidt equal-area projection.

(McElhinny, 1966a), while Jones and McElhinny (1966), using the palaeomagnetic method, correlated post-Waterberg diabases in Botswana and the then Transvaal Province of South Africa with the UD. Later, McDonald and Andersen (1973) found that gabbroic intrusions in the north of the then Orange Free State were part of the same episode of igneous activity—the remanence was however reversed relative to that in the UD and its correlatives. Wilson et al. (1987) recorded an UD direction in a dyke some 200 km west of the principal UD outcrops, while Mushayandebvu et al. (1994) found evidence that the remanence of the Umvimeela Dyke, west of its parent Great Dyke, contained a component identifiable as an UD overprint. Lastly, Hargraves et al. (1994) showed that the Timbavati gabbros in the eastern Transvaal (now Mpumalanga) were part of the same episode. All of the above rocks are now regarded as being part of the Umkondo Large Igneous Province (ULIP) and there are 44 widespread palaeomagnetic sites giving a mean pole (UMK) position at 65.4°N and 37.2°E ($K = 40.8$, $A_{95} = 3.4°$). Thirty-seven of the virtual geomagnetic poles (VGP) are south poles (i.e., with reverse polarity); the remaining seven, north poles (normal polarity), are all from sites in the most southerly part of the ULIP. The characteristic remanence observed in the basic ULIP rocks is believed to be primary even though little rock magnetic experiments have been performed on them. There is a positive baked contact test involving the post-Water-

berg diabase sill that cuts the Premier Mine kimberlite (Jones and McElhinny, 1966; Jones, 1968). Although not a full test (no zone of overlapping country rock/sill remanence was observed), the results support a primary origin for the sill's remanence. The sites are spread throughout the region and involve different rock types; flat-lying, undeformed sediments are often the hosts for the intrusive rocks; many sites are situated hundreds of kilometres from rocks deformed and metamorphosed in later events, e.g., the Pan-African, and there are older formations throughout the region that have retained their primary remanence (McElhinny and Opdyke, 1964).

K–Ar ages from the UD and equivalents reported by Jones and McElhinny (1966) ranged from 590 to 1136 Ma with the low ages attributed to argon loss. Allsopp et al. (1967) dated the post-Waterberg-diabase sill cutting the Premier Mine kimberlite at 1115 ± 15 Ma (1090 Ma using a Rb–Sr decay constant of 1.42×10^{-11} year^{-1}) and this result agrees with the Rb–Sr result (1080 + 140/ − 25 Ma) of Allsopp et al. (1989) from a 47 point error chron and 3 separated biotite analyses. Hanson et al. (1998) reported a U–Pb zircon age of 1105 ± 2 Ma of a granophyre horizon within an Umkondo sill in southeastern Zimbabwe, an age confirmed by Wingate (2001) (1099 ± 9 Ma; U–Pb SHRIMP age on baddelyite and zircon) from samples from an Umkondo sill in northeastern Zimbabwe. The time of emplacement for the Umkondo dolerites and their correlatives is thus considered to be ca. 1100 Ma.

The agreement between the BM and the UMK pole from the ULIP is a strong evidence that the Grunehogna craton and the Kalahari craton were in close conjunction some 1100 Ma ago. One polarity only has been observed in the Borgmassivet rocks, which is consistent with the dominant "reversed" polarity in northern Kalahari. However, there is an apparent ca. 30–20 Ma age difference between the two rock units. Whether or not such a difference is real awaits future geochronological work, particularly in the Borgmassivet rocks.

However, the ca. 1110 Ma CL pole from the Coats Land (Gose et al., 1997) falls significantly away from the BM pole, and indeed from the entire apparent polar wander path (APWP) of the Kalahari craton for the interval of ca. 1165–1000 Ma (for details of Kalahari poles other than the UMK pole, see Powell et al., 2001, Table 1). Gose et al. (1997) recognised this discrepancy, and suggested that the difference between the CL and BM poles is likely due to their age difference. However, we question the validity of such interpretation because even if we bring the CL pole closer to the Kalahari APWP by rotating the Maud Province (and its two palaeomagnetic poles) 20° around a local vertical axis (see arrows from the two poles in Fig. 8), the younging direction between the two poles still opposes that of the Kalahari APWP, and the CL pole would still fall outside of the APWP.

Accepting the reliability of the CL pole and its age, a more likely interpretation for the discrepancy between the two Maud Province poles is possible tectonic movement between Coats Land and the Grunehogna craton since ca. 1110 Ma, either during the Grenville-age event, or the Pan-African events (e.g., Jacobs et al., 1998).

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